

Requirement Document

The **Restaurant Management System** is designed to automate and simplify the core operations of a restaurant by providing a centralized database management solution. The system manages essential business entities including employees, roles, menu categories, menu items, customers, restaurant tables, and customer orders. The database is designed following relational database principles to ensure data integrity, minimize redundancy, and support efficient data retrieval and management.

1 - The restaurant has a number of employees Each employee has **EmployeeID** (unique identifier), FirstName, LastName, PhoneNumber, Gender, HireDate, Salary, JobTitle, Username, and Password.

2 - Each employee must belong to only one role. Each role has **RoleID** (unique identifier) and RoleName (Admin, Cashier, Waiter). One role may be assigned to several employees.

3 - The restaurant manages different categories of menu items. Each category has **CategoryID** (unique identifier) and CategoryName. Each category may contain several menu items, while each menu item belongs to only one category.

4 - The restaurant offers a variety of menu items. Each menu item has **ItemID** (unique identifier), ItemName, Description, Price, and Availability Status. Each menu item belongs to one category.

5 - The restaurant serves different customers. Each customer has **CustomerID** (unique identifier), FullName, and PhoneNumber. One customer may place several orders.

6 - The restaurant contains several dining tables. Each table has **TableID** (unique identifier), TableNumber, Capacity, and Status (Available or Occupied). One table may be assigned to several orders over time, while each order is assigned to only one table.

7 - The restaurant records customer orders. Each order has **OrderID** (unique identifier), OrderDate, OrderStatus, and TotalAmount. Each order is created by one employee (Cashier or Waiter), belongs to one customer, and is assigned to one restaurant table. One employee, one customer, or one restaurant table may be associated with multiple orders.

8 - Each order may contain multiple menu items, and each menu item may appear in multiple orders. This many-to-many relationship is implemented through the **OrderDetails** table, which stores the **OrderID**, **ItemID**, **Quantity**, and **UnitPrice**, with **OrderID** and **ItemID** forming the composite primary key.